CIS 181: Object-Oriented Programming II

Project #2

Binary Decision Trees, Interactive Text-based Game! (85 points total)

Objectives

- To gain experience with using recursion by studying recursive methods: *recursiveLoad, recursiveSave, and recursiveBuildTree*.
- To gain experience with implementing and using traversing reference-based structures.
- To gain experience analyzing existing code and tracing through it to understand how it works.

Description

This is a solo project. You may not work with anyone else. You may get help from the TA's, mentors, and instructor.

The game is simple. You are posed with a situation, to which you are given two options. One option leads the story one way, the other, a different. The story eventually ends any which way you choose. Here are a couple of examples to demonstrate:

```
You encounter a wolf, do you? 1 = Fight the Wolf 2 = Run Away
2
The wolf is in hot pursuit, do you? 1 = Jump in the river 2 = Climb up a tree
2
You made it up the tree, after a while the wolf gets bored and leaves. You are safe,
you win!
```

```
You encounter a wolf, do you? 1 = Fight the Wolf 2 = Run Away
2
The wolf is in hot pursuit, do you? 1 = Jump in the river 2 = Climb up a tree
1
You can't swim, you drowned!  Game Over!
Would you like to play again?
n
```

How does it work? We are using the concept of a binary tree to make decisions. The binary trees work similar to doubly linked lists, in that the they have 2 sets of reference pointers. Instead of the nodes being in one linear “train”, the reference points (left and right) branch off to another level, one to the left, one to the right. In our case, a left and right of any node will never loop back to a previous nodes level. I have provided code that will help you build the tree, but you will have to analyze how it does this. Be forewarned, it is recursive!

Exercises

1. Download the following class source files:
   - DevisionTree.java
   - TextDecisionGame.java
   - TextDecisionGameBuilder.java
   - TreeNode.java

2. In Eclipse, create a new Java project called "Project 2", and import the above 3 files.

3. Both TextDecisionGameBuilder.java and TextDecisionGame.java have main methods. TextDecisionGameBuilder is the one that will help you to build your games. This will compile and correctly run without further modification.

4. Using TextDecisionGameBuilder, generate 3 of your own text-based games. Each game should have at least one path to completion that is at least 5 scenarios deep. (15 points)

5. Analysis the recursive methods recursiveBuildTree, recursiveSave and recursiveLoad in the class DecisionTree and gain an understanding of how the trees are is built. Generate a visual representation in tree form of the order in which each node is created. Put a number inside the node to denote the order (you do not need to put the scenarios in). (20 points)

6. TextDecisionGame is the second program with main, and this is the actual “game” launcher. When you run it, it will ask for a save file you created with TextDecisionGameBuilder. In order for it to work, you need to implement the method play() inside the DecisionTree class. This method is NOT recursive. You will use a loop, and ask the user what option they wish to choose… moving a traverse pointer to the next node in the tree based on the users choice. (example: trav=trav.getOption1() or trav=trav.getOption2(). (45 points)

7. Make sure you comment your program. This includes name, date, description, as well as general comments, particularly in the play method.. (5 points)

8. Submit your completed program and scans/pictures to myCourses. Any documents (including this description pdf), source code, or work you create/modify as a result of this project is the property of the University of Massachusetts Dartmouth. This document and any and all source code cannot be shared with anyone except: University of Massachusetts Dartmouth faculty (including TA's), and in a private digital portfolio (public access online is prohibited) with the intention of applying to jobs and internships. These exceptions are non-transferable. Failure to comply is, at the very least, an academic infraction that could result in dismissal from the university. Your submission of this project to myCourses is your acknowledgement of these stipulations.
